Yunzhi Lin

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Education

University of California, Berkeley *GPA: 3.44/4.00*

B.S. Bioengineering *Expected Graduation: May 2028*

Coursework: Electronics for IoT, Mechanical Behavior of Materials, Polymeric Materials, Properties of Materials, Biomechanics, Computational Biology, Biochemistry, Tissue Engineering, Structure and Interpretation of Computer Programs

Skills

Software: SolidWorks, CREO, Windchill, Fusion 360, OnShape, 3DExperience, FEBio, Ansys, Python, Java, SQL, Git, Scheme
Techniques: DFMA, GD&T, 3D Printing, Bioprinting, Lean Manufacturing, Machine Maintenance and Improvement,
Good Manufacturing Practice (GMP), Good Documentation Practice (GDP)
Awards: Bakar Ignite Scholar; 2nd Place (California) in Mobile Robotics Technology of the 2023 SkillsUSA Championship

Experience

GE HealthCare *Waukesha, WI*

Mechanical Engineering Intern - Maternal Infant Care *01/2026 – 07/2026*

- Designed 13 production-ready GD&T drawings for sheet metal and injection-molded parts in Creo, performed tolerance stack-up analysis, Ansys FEA, and DFMA reviews to support NPI electromechanical integration and manufacturing readiness.
- Assessed suspended-mass safety risks across 90+ components using energy calculations and FMEA, implementing secondary retention, fail-safe mechanisms, and poka-yoke design features to improve compliance with medical device safety standards.
- Designed test fixtures, developed and executed test plans, and analyzed data for thermal management, center-of-gravity, sensor, and acoustic testing to support design verification and validation in compliance with ISO, IEC 60601, and system requirements.
- Co-authored a defensive publication proposing AI-integrated tool suspended-mass analysis tool to improve workflow efficiency.

Silver Lake Research Corporation *Irwindale, CA*

Engineering Associate *09/2024 – 08/2025*

- Authored and updated comprehensive SOPs for operation, maintenance, and calibration of 5 manufacturing machines, standardizing procedures to align with GMP and improve training, reliability, and audit readiness.
- Developed a scheduling system to centralize task assignments across Manufacturing, Production, and QC, improving real-time task visibility, reducing scheduling conflicts, and accelerating project handoffs.
- Reduced end-to-end production lead time by 18% and minimized material waste by leading root-cause analyses on recurring defects and executing targeted equipment and process modifications.
- Designed custom machine fixtures in OnShape, prototyped them with 3D printing, and applied GD&T to improve machine usability and production repeatability.

Engineering Intern *05/2024 – 08/2024*

- Authored and executed 7 validation protocols for new manufacturing machines and software tools, completing 7 verification and validation reports to confirm system performance under varying constraints and enhanced QC efficiency.
- Developed a cost analysis and production capacity evaluation tool in Excel for 4 products, integrating Macros and VBA Scripts to allow testing of different constraints and processes, resulting in more cost-efficient production workflows.
- Designed a 3D production layout for assemblies using Creo, optimizing space utilization, material flow, and environmental control, leading to increased operational efficiency.
- Led maintenance and troubleshooting on key manufacturing machines, enhancing reliability and reducing downtime, while honing technical skills in equipment management.

UC Berkeley Mechanical Engineering *Berkeley, CA*

Undergraduate Research Assistant — advised by Professor Grace Gu *02/2025 – 12/2025*

- Fabricated collagen-alginate 3D-bioprinted composite scaffolds across multiple blend ratios to emulate ligament mechanics and evaluated material properties through tensile and compression testing.
- Optimized printing and post-processing parameters to improve strength, shape stability, and durability of scaffold prototypes.
- *Tuning collagen and collagen-alginate mechanics through extrusion bioprinting process parameters*, RSC Advances, 2025.

Projects

IoT Electromechanical Marble Maze *10/2025 – 12/2025*

- Developed a networked mechatronic maze with PID-controlled 2-axis tilt, 4 randomized PWM servo gates, and sensor-driven game logic, integrating ESP32 MicroPython firmware and validating closed-loop behavior through iterative testing and tuning.
- Designed and fabricated the mechanical system using Fusion 360, 3D printing, and laser cutting, and implemented ESP-NOW wireless communication and HTTP-based game data logging to a Google Apps Script-hosted live leaderboard.